

Article

The Sensitivity Analysis of Parameters in the 1D–2D Coupled Model for Urban Flooding

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Abstract: The ongoing changes in climate and the rapid pace of urbanization are contributing to an alarming increase in the prevalence of urban flooding, which is having a profound impact on the quality of life for residents and the smooth functioning of urban areas. The 1D–2D coupled model is an effective tool for simulating the process of urban flooding, thereby providing a scientific basis for urban planning, flood prevention, and mitigation strategies. The values of numerous parameters within the model not only influence the computational efficiency but also influence the precision of the simulation outcomes. It is of particular significance to ascertain the sensitivity of model parameters. In this study, a 1D–2D coupled model of urban flooding was constructed, and a parameter sensitivity analysis was conducted using the modified Morris method and the Sobol method in two ways, with the amount of waterlogging as the target. The findings indicate that the minimum infiltration rate is the most sensitive parameter in the local sensitivity analysis, whereas the Manning coefficient of the permeable surface area is the most sensitive in the global sensitivity analysis. The research outcomes can facilitate the optimization of the model parameters and enhance the precision and dependability of the model predictions, thereby providing more accurate data support for urban flooding early warning and emergency response.

Keywords: coupled model; urban waterlogging; sensitivity analysis; water volume



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1. Introduction

In recent years, with the acceleration of global climate change and urbanization, the frequent occurrence of heavy rainfall events has led to increasingly severe urban waterlogging problems [1], which has placed higher demands on urban drainage systems. The effective response to heavy rainfall and flooding disasters requires the establishment of accurate hydrological and hydrodynamic models for a variety of urban waterlogging studies, such as urban drainage capacity analysis [2], flood inundation process analysis [3], waterlogging causes analysis [4], waterlogging risk map preparation [5], waterlogging risk assessment [6], and so on. How to accurately simulate the formation of urban flooding and waterlogging processes under heavy rainfall conditions has become the direction of urban flooding research. The SWMM (Storm Water Runoff Model) and Telemac (Hydrodynamic Model) are currently the two most widely used tools in urban drainage modeling. The SWMM is mainly used to simulate the rainfall runoff process in the city [7]. It is widely used in drainage system design, storm water flood management, and pollutant transport

simulation, and its limitation is that it can only simulate the water movement within the pipe network; it cannot accurately deal with the surface water and its diffusion and flow in the complex terrain and ignores the dynamic interaction between the urban surface and the pipe network system. Telemac [8], on the other hand, with its advanced hydrodynamic computational capability, is able to simulate complex two-dimensional water movement and water accumulation phenomena, which is lacking in the simulation of water flow in underground pipe networks. The coupling of these two models can make full use of the SWMM's efficient simulation capability of pipe networks and Telemac's fine description of the complex flow on the surface to realize the comprehensive simulation of the interaction of water flow between the surface and pipe networks and improve the model's overall accuracy. The accuracy of the urban flood model is largely dependent on the parameter values, which often have large uncertainties, and parameter sensitivity analysis is an important method to ensure the reliability of the model [9]; through sensitivity analysis, the key parameters that most significantly affect the model output can be identified [10] so as to improve the efficiency of the model rate determination.

Sensitivity analysis plays a crucial role in urban flooding simulations by identifying the most influential parameters affecting model performance. It is primarily categorized into local sensitivity analysis and global sensitivity analysis. Local sensitivity analysis provides a quick assessment of parameter influence by perturbing one variable at a time while keeping others constant, whereas global sensitivity analysis evaluates parameter effects across the entire input space and considers interactions between multiple parameters. Recent research has employed both approaches in urban flood modeling, utilizing methods such as the Morris method for screening and the Sobol method for detailed variance decomposition. The present study builds upon this research by integrating the Morris and Sobol methods, leveraging the efficiency of Morris for screening and the accuracy of Sobol for a deeper understanding of parameter interactions.

Several studies have applied local sensitivity analysis to assess key hydrological parameters in urban flooding models. Huang Jinliang et al. [11] investigated rainfall–runoff dynamics in Macao, concluding that imperviousness is the most sensitive factor influencing runoff depth. Similarly, Gao Yinghui et al. [12] found that imperviousness and minimum infiltration rate significantly impacted runoff, while diffuse flow width also played a crucial role in determining runoff coefficients. Tan Minghao et al. [13] employed the Storm Water Management Model to analyze pollutant washout and identified land-use parameters as key drivers of water quality fluctuations.

In addition to surface hydrology, Zeng Jiajun et al. [14] examined flood modeling in Guangzhou Tianhe Smart City and highlighted maximum infiltration rate and surface Manning's coefficient as the most influential parameters. Jin Mengxiao et al. [15] extended this analysis to low-impact development facilities in the Storm Water Management Model, revealing that bioretention pond conductivity and plant cover significantly affected peak flow rates. Jiang et al. [16] further explored parameter sensitivity in the Storm Water Management Model, demonstrating that hydrological and water quality parameters exhibit significant sensitivity in urban runoff modeling.

While local sensitivity analysis is computationally efficient and ideal for preliminary parameter screening, its primary limitation is its inability to capture interactions among parameters, making it unsuitable for comprehensive flood model optimization. However, it remains valuable for identifying the most influential single parameter in models where dominant factors drive system behavior.

Unlike local sensitivity analysis, global sensitivity analysis provides a more comprehensive assessment by quantifying both individual and interactive parameter effects. This makes it well suited for evaluating complex urban flood models where multiple vari-

ables contribute to system behavior. The most commonly used global sensitivity analysis methods include the Sobol method, Mutual Information method, Generalized Likelihood Uncertainty Estimation method, and Enhanced Fourier Amplitude Sensitivity Test method.

Chang et al. [17] applied the Sobol method to the Storm Water Management Model simulations, successfully identifying key parameters that drive urban hydrological processes. Lin Zhang et al. [18] refined the Sobol variance decomposition method, achieving more consistent sensitivity rankings and improving parameter reliability. Liao Ruting et al. [19] utilized the Mutual Information method to analyze sensitivity in the Storm Water Management Model, revealing that Manning's coefficient of permeable areas and minimum infiltration rate were the most significant factors influencing flood flow and runoff coefficients.

More advanced global sensitivity analysis applications have incorporated hybrid methodologies. Tang Qingzhu et al. [20] combined the Generalized Likelihood Uncertainty Estimation method and a modified Morris method to examine urban flood model sensitivity in Jinan, highlighting channel roughness and attenuation coefficient as the most sensitive parameters. Wu et al. [9] investigated the effects of calibration event numbers on model performance, integrating BPROME and the Generalized Likelihood Uncertainty Estimation method to determine that 3–5 calibration events optimized accuracy while minimizing data collection costs.

Furthermore, studies by Dobson et al. [21] and Chenlei et al. [22] explored variance-based sensitivity analysis in synthetic urban drainage models and FLOWS-Tank models, respectively, emphasizing the importance of water level fluctuations in urban flood risk assessments. Xu et al. [23] extended global sensitivity analysis applications to dynamic flood risk analysis, highlighting the role of parameter interactions under future uncertainty scenarios.

While global sensitivity analysis offers a more detailed analysis compared to local sensitivity analysis, it is computationally demanding, particularly for high-dimensional parameter spaces. The Sobol method, despite its accuracy, requires an extensive number of model simulations, making it computationally expensive for real-time flood risk assessments.

Recent research has shown that no single sensitivity analysis method is universally optimal for urban flood modeling. The Morris method provides a computationally efficient screening tool, identifying non-influential parameters while maintaining low computational costs. The Sobol method, in contrast, quantifies variance contributions and detects higher-order interactions, making it ideal for comprehensive assessments.

Studies have demonstrated the advantages of combining both approaches. Xu et al. [23] showed that Morris–Sobol integration enhances flood model efficiency, allowing for rapid initial screening, followed by detailed sensitivity quantification. Peiris et al. [24] further validated this approach in urban flood event analysis, highlighting its superior performance in evaluating complex hydrological interactions. Jiang et al. [16] and Dobson et al. [21] also provided supporting evidence that Morris–Sobol methods improve computational feasibility while maintaining analytical depth.

Compared to existing studies that apply only the Morris or Sobol methods, the present study builds upon this research by demonstrating that their combined application provides a more balanced approach. Previous studies utilizing Morris alone often lack interaction assessments, while studies relying solely on Sobol require high computational costs. By integrating Morris and Sobol, this study maximizes efficiency while retaining analytical rigor, making it a robust method for urban flood model optimization.

While there is a substantial body of literature on sensitivity analysis, the majority of studies have focused on one-dimensional models, with relatively few sensitivity analyses

conducted on coupled models. This paper employs the Morris method and the Sobol method to perform sensitivity analyses on 12 parameters of the SWMM. It does so with the objective of exploring the influence mechanisms of the parameters on the amount of water accumulation under different rainfall conditions. The Morris method is used to quickly screen important parameters with high computational efficiency, while the Sobol method is employed to reveal the interaction and nonlinear effects of parameters through global analysis. By integrating these two methods, one can achieve a balance between efficiency and accuracy, ensure the reliability of the screening results, and provide support for subsequent in-depth analysis, thereby establishing a more accurate data foundation for early warning of urban flooding.

2. Method

2.1. Construction of the Coupled Model

2.1.1. Overview of the SWMM 1D Model

The SWMM (Version 5.2.4) is a hydrologic and hydrodynamic model that calculates various hydrological processes, including infiltration, evapotranspiration, runoff, ponding, and sewer conveyance, in response to rainfall events. It comprises two main modules: a hydrologic module for calculating rainfall runoff processes for different surface types in the watershed and a hydraulic module for modeling water movement in drainage facilities, such as pipes and open channels.

The Horton model is commonly used to describe the infiltration process. The equation is as follows:

$$i(t) = i_c + (i_0 - i_c)\exp(-kt) \quad (1)$$

where $i(t)$ is the infiltration rate at time t , i_0 is the initial infiltration rate, i_c is the constant (or final) infiltration rate, and k is the decay constant.

The flow in drainage pipes is typically computed using Manning's equation, which is given by the following:

$$Q = \frac{1}{n} A R^{\frac{2}{3}} S^{\frac{1}{2}} \quad (2)$$

where Q is the flow rate, n is Manning's roughness coefficient, A is the cross-sectional area of the conduit, R is the hydraulic radius, and S is the slope of the channel.

2.1.2. Overview of the Telemac 2D Model

Telemac (Version 8.5) was developed by Électricité de France (EDF) in its Laboratoire National d'Hydraulique et Environnement (LNHE) in the 1990s and has since evolved into an open-source, multifunctional hydrodynamic simulation software. Telemac employs the finite element or finite volume method to solve the shallow water equations by generalizing the surface of the study area to multiple triangular grids and performing water and momentum balance calculations for each grid.

In its computational process, Telemac primarily involves the continuity equation and the momentum equations.

The 2D continuity equation is written as follows:

$$\frac{\partial h}{\partial t} + \frac{\partial(uh)}{\partial x} + \frac{\partial(vh)}{\partial y} = 0 \quad (3)$$

where h is the water depth and u and v are the horizontal velocity components in the x and y directions, respectively.

The momentum equations in the x and y directions are expressed as follows:

$$\frac{\partial(uh)}{\partial t} + \frac{\partial(u^2h)}{\partial x} + \frac{\partial(uvh)}{\partial y} + gh\frac{\partial h}{\partial x} = -ghS_{fx} + \text{viscous terms} \quad (4)$$

$$\frac{\partial(vh)}{\partial t} + \frac{\partial(uvh)}{\partial x} + \frac{\partial(v^2h)}{\partial y} + gh\frac{\partial h}{\partial y} = -ghS_{fy} + \text{viscous terms} \quad (5)$$

where g is the gravitational acceleration, S_{fx} and S_{fy} are the friction slopes in the x and y directions, and “viscous terms” represent the diffusion effects and may include turbulence closure models (such as the $k - \varepsilon$ model) to account for turbulence.

2.1.3. Coupling of 1D and 2D Models and Parameter Configuration

The SWMM has a complete hydrological hydrodynamic module, which can exhaustively simulate the flow of water in the pipe network but cannot complete the simulation of the urban waterlogging diffusion process in the two-dimensional surface, and Telemac does not have an urban pipe network module but can complete the simulation of urban waterlogging diffusion in the surface; therefore, the coupling of the two can complete the completion of urban waterlogging from the production of sinks, pipe network hydrodynamics, and the completion of urban surface hydrodynamic simulation, that is, extract the SWMM pipe network overflow process flow and input into Telemac, the latter to complete the diffusion of urban waterlogging in the surface.

The SWMM adopts the power wave method to complete the simulation of the pipe network; the Horton model is adopted for the calculation of production and sink flow; Telemac adopts the finite volume method to calculate; in order to ensure the stability of the calculation, a variable step size is selected when generating the two-dimensional surface mesh in order to take into account the accuracy and efficiency; a 30 m mesh is selected; for the main area to adopt a local encryption, a 10 m mesh is selected; the initial depth of surface water is 0; and the research zone boundary is open boundary.

2.2. Parameter Sensitivity Analysis Method

2.2.1. Parameter Selection

There are 13 parameters in the SWMM, of which catchment area, imperviousness, slope, width, and pipe length can be directly determined by measurement and analysis. Some data of subcatchments, such as area and slope, are obtained by processing DEM data in ArcGIS. These parameters are measured values rather than empirically assigned ones. The remaining 8 parameters in the SWMM are empirical, and Telemac uses 4 empirical parameters for urban waterlogging simulation. All empirical parameter names and value ranges are determined based on modeling manuals and literature research, as shown in Table 1.

Table 1. Names and value ranges of model parameters.

Model	Parameter	Description	Unit	Minimum Value	Maximum Value
SWMM	N-Imperv	Manning’s Coefficient for Impervious Areas	/	0.01	0.024
	N-Perv	Manning’s Coefficient for Pervious Areas	/	0.01	0.08
	S-Imperv	Depression Storage for Impervious Areas	mm	1.3	3
	S-Perv	Depression Storage for Pervious Areas	mm	3	8
	Maxrate	Maximum Infiltration Rate	mm/h	50	100
	Minrate	Minimum Infiltration Rate	mm/h	2	30
	Dc	Decay Coefficient	/	2	7
	Roughness	Conduit Roughness	/	0.011	0.017

Table 1. Cont.

Model	Parameter	Description	Unit	Minimum Value	Maximum Value
Telemac	Friction	Surface Friction	/	0.01	0.8
	Fc	Stable Infiltration Rate	mm/h	0	100
	Vd	Velocity Diffusion Coefficient	/	1.00×10^{-6}	1.00×10^{-3}
	CFL	Courant Number	/	0.2	0.9

2.2.2. Parameter Sensitivity Analysis

1. Local Sensitivity Analysis

The Morris method [25], also known as the one-at-a-time (OAT) method, is a local sensitivity analysis approach first proposed by Morris in 1991. It evaluates the influence of each input parameter on model outputs by sequentially altering the input parameters and observing the resulting changes in the output. The Morris method determines parameter sensitivity by calculating the impact of each parameter's variation. The sensitivity E_i of a parameter is calculated using the following formula:

$$E_i = \frac{e_a - e_b}{\Delta x_i} \quad (6)$$

In the formula, e_a represents the accumulated water volume in the model output after the parameter change, e_b represents the accumulated water volume in the model output before the parameter change, and Δx_i is the magnitude of the variation in the analyzed parameter Δx_i .

The Morris method employs the mean value and the absolute mean value as criteria for determining sensitivity. The formulas are as follows:

$$\mu_i = \frac{1}{r} \sum_{j=1}^r E_i \quad (7)$$

In the formula, μ_i represents the overall strength of the parameter's influence on the output. A larger mean value indicates that the parameter has a greater average impact on the model output. r represents the number of sampling iterations.

However, since the sign of μ_i reflects whether the perturbation has a positive or negative impact on the output, the calculated sensitivity μ_i might underestimate the parameter's influence if its effects alternate between positive and negative (i.e., sometimes increasing the output and other times decreasing it), resulting in a cancellation effect. To eliminate this cancellation effect, μ_i^* , the absolute mean value, is introduced to measure parameter sensitivity. Regardless of whether the sensitivity is positive or negative, μ_i^* captures the total influence of the parameter. This adjustment makes the analysis more precise, particularly when a parameter causes significant fluctuations in the model output. μ_i^* better reflects the parameter's impact in such cases.

$$\mu_i^* = \frac{1}{r} \sum_{j=1}^r |E_i| \quad (8)$$

In the formula, μ_i^* represents the absolute mean sensitivity of the i -th input parameter.

2. Global Sensitivity Analysis

The Sobol method is based on the principle of variance decomposition, which partitions the total variance of the model output into contributions from different input variables

and their combinations. For a model with k input variables $X_1, X_2, \dots, X_k$, where the output is $Y = f(X_1, X_2, \dots, X_k)$, the total variance of the model output can be decomposed as follows:

$$\text{Var}(Y) = \sum_{i=1}^k V_i + \sum_{i < j} V_{ij} + \sum_{i < j < l} V_{ijl} + \dots + V_{12\dots k} \quad (9)$$

In the formula, V_i represents the first-order sensitivity of input variable X_i , indicating the contribution of X_i alone to the output variance, and V_{ij} represents the second-order sensitivity, reflecting the contribution of the interaction between X_i and X_j to the output variance. Similarly, $V_{12\dots k}$ represents the joint sensitivity of all input variables.

S_1 is the first-order sensitivity index in the Sobol method. It indicates the proportion of the total variance contributed by a single parameter X_i to the model output variance. To measure the independent impact of this parameter on the output, the formula for S_1 is as follows:

$$S_1 = \frac{V_i}{V} = \frac{\text{Var}[E(Y|X_i)]}{\text{Var}(Y)} \quad (10)$$

In the formula, S_1 represents the first-order sensitivity index of the input parameter X_i , and $\text{Var}[E(Y|X_i)]$ denotes the variance of the conditional expectation of the model output Y given X_i , while other parameters are held constant. This reflects the contribution of X_i to the uncertainty of the output.

To measure the overall impact of the input variable X_i , including interactions, on the output, the total sensitivity index S_T in the Sobol method is introduced. S_T represents the contribution of X_i , accounting for all interactions, to the variance of the model output. This includes both the direct effect of the parameter and its interactive sensitivity with other parameters. The formula for calculating S_T is as follows:

$$S_T = 1 - \frac{\text{Var}[E(Y|X_{\sim i})]}{\text{Var}(Y)} \quad (11)$$

In the formula, S_T represents the total sensitivity index of the input parameter X_i . $\text{Var}[E(Y|X_{\sim i})]$ is the variance of the conditional expectation of the output Y when all parameters except X_i are held constant.

The S_T value reflects the total contribution of parameter X_i , including its direct effect and its interactions with other parameters. A large S_T value indicates that the parameter and its interactions with other parameters have a significant impact on the output variance.

2.3. Identification of Rainfall Pattern

According to the latest research results [26], two extreme heavy rainfall patterns and one Chicago rainfall pattern (Chicago50a) are selected, among which the most unfavorable rainfall pattern of the city (szzbl50a) has a great impact on the city infrastructure, and the most frequent rainfall pattern (cxzd50a) is one of the most typical rainfall patterns in Nanchang. Currently, the flood protection standard of Nanchang City is one in 50 years, so the 50a rainfall data of the three rainfall patterns are selected as model inputs, and the three rainfall patterns are shown in Figure 1.

The design storm formula for Nanchang City is as follows:

$$q = \frac{1598 \times (1 + 0.69 \log P)}{(t + 1.4)^{0.64}} \quad (12)$$

where q is rainfall intensity (mm/h or L/(s·ha), P is return period (years), and t is rainfall duration (minutes).

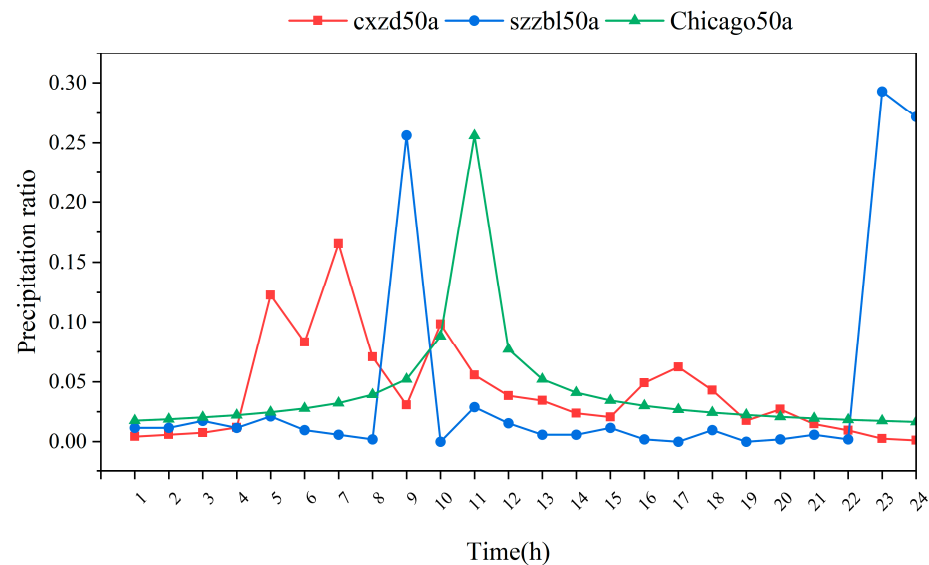


Figure 1. Identification of rainfall patterns with long durations in Nanchang City.

3. Case Study

3.1. Overview of the Study Area

Nanchang City is located in the southern part of the northern part of Jiangxi Province, with geographic coordinates of $28^{\circ}41' \sim 29^{\circ}11'$ N latitude and $115^{\circ}27' \sim 116^{\circ}35'$ E longitude, which belongs to a typical subtropical humid monsoon climate zone. The climate of this region is characterized by a remarkable average annual precipitation of about 1611.8 mm, frequent heavy rainfall, and an average number of days of heavy rainfall of 5.7 days per year, which is mainly affected by the Jianghuai Meiyu Frontal rainstorms [27]. The JAC Meiyu Feng rainstorms usually occur from May to July, with long duration and high rainfall intensity, often accompanied by thunderstorms and short-term heavy precipitation, which is very likely to cause flooding. In this paper, Honggutan District of Nanchang City is selected as the research object, which is located on the right bank of Ganjiang River and belongs to one of the central urban areas of Nanchang City, which is an important area for the rapid development of the city in recent years. The catchment area of Honggutan District is 10 square kilometers, and the surface elevation ranges from 15.6 m to 29.5 m. As shown in Figure 2. The terrain is generally gently sloping, and there are low-lying areas in some areas, which tend to form waterlogged areas during heavy rainfall. In addition, the high density of development in the county and the extensive distribution of roads, buildings, and other impervious surfaces further exacerbate the rapid accumulation of surface runoff and the risk of waterlogging.

3.2. Data Sources and Processing

The DEM data used in this study have an accuracy of 30 m and were obtained from the Geospatial Data Cloud Platform of Chinese Academy of Sciences, the land use data were obtained from Wuhan University, and the pipe network data were obtained from Nanchang Urban Planning and Design Research Institute. After processing the above information, the model of Honggutan district was built in the SWMM and Telemac, respectively, and the coupling mode was loose coupling. A total of 641 stormwater inspection wells, 629 pipes, and 622 catchment areas were generalized in the SWMM, and the generalization diagram is shown in Figure 3a; a total of 238,085 two-dimensional meshes and 119,863 mesh nodes were generated in Telemac, and the generalization diagram is shown in Figure 3b.

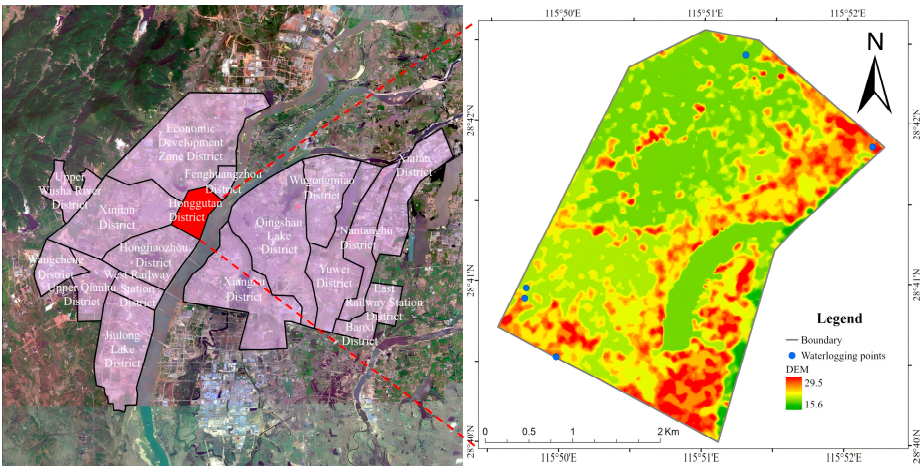


Figure 2. Geographic location of Honggutan District and distribution of existing waterlogging points.

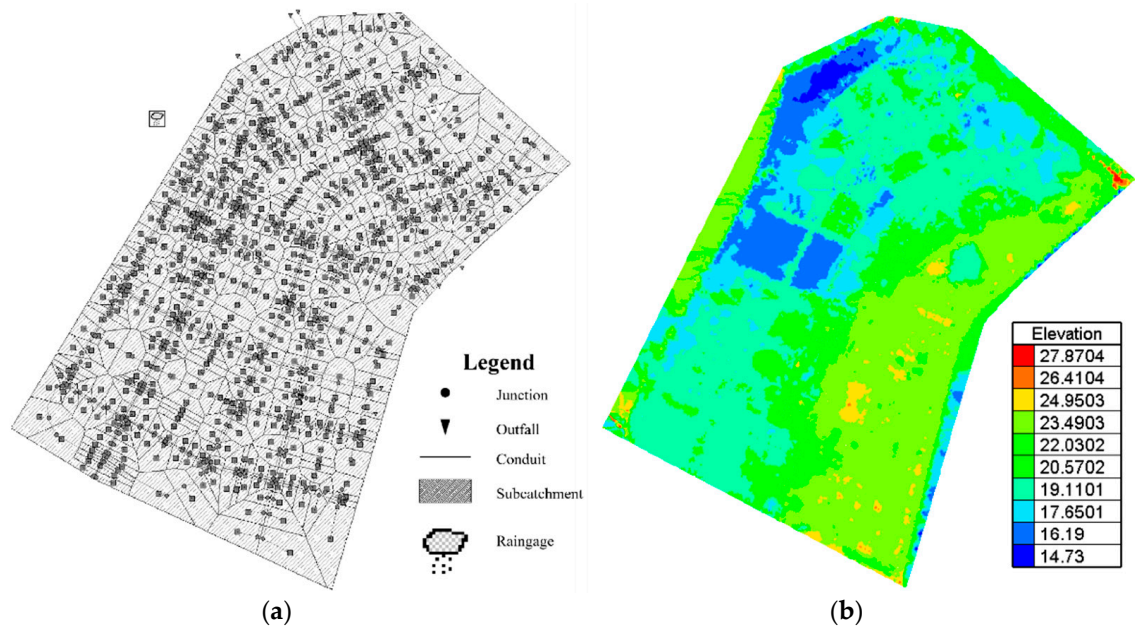


Figure 3. Simplified coupled model diagrams: (a) simplified SWMM diagram; (b) simplified Telemac diagram.

3.3. Coupled Model Validation and Calibration

Four rainfall events (20230608, 20220610, 20220629, 20210825) are selected as inputs to simulate the waterlogging process of the coupled model, and the rate is determined and verified based on the measured waterlogging points. The measured waterlogging points are completely overlapped with the simulated waterlogging points, the model has a certain degree of reliability, the values of the parameters after the rate is determined are shown in Table 2, and the subsequent sensitivity analysis is based on this result.

Table 2. Calibration results of the coupled model parameters.

Parameter	N-Imperv	N-Perv	S-Imperv	S-Perv	Maxrate	Minrate
Value	0.018	0.015	2.4	7	80	25
Parameter	Dc	Roughness	Friction	Fc	Vd	CFL
Value	2	0.015	0.2	50	1×10^{-6}	0.4

4. Results and Analysis

4.1. Local Sensitivity Analysis

By analyzing the sensitivity of the Morris method parameters under different rainfall patterns (cxzd50a, szzbl50a, and Chicago50a), positive values indicate that an increase in the value of the parameter has a positive effect on water volume, and negative values have the opposite effect, as shown in Figure 4a. Minimum infiltration rate (Minrate) showed the most significant negative effect in all three rainfall patterns, with sensitivity coefficients of -0.537 (cxzd50a), -0.131 (szzbl50a), and -0.448 (Chicago50a), indicating that increasing this parameter significantly reduced the water volume. Pipe roughness (Roughness) showed a positive correlation in all three rainfall patterns, and its sensitivity coefficient was very close to 0.396 and 0.395 for the cxzd50a and Chicago50a rainfall patterns, respectively, while it was 0.270 for the szzbl50a rainfall pattern, indicating that higher roughness increases the water volume.

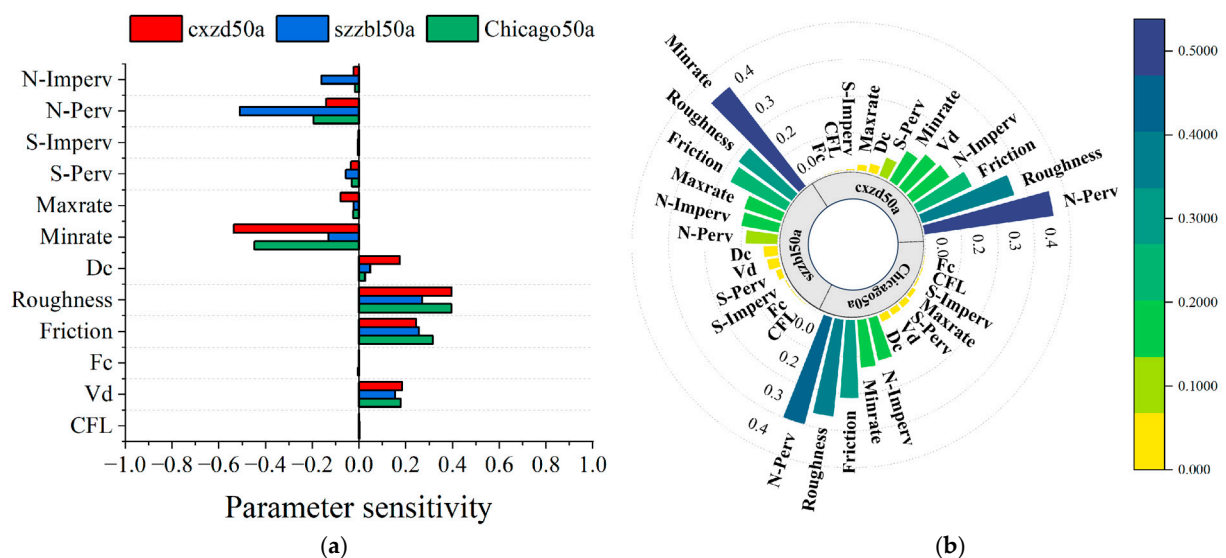


Figure 4. Morris method sensitivity analysis. (a) Sensitivity μ_i of different parameters; (b) Sensitivity strength ranking μ_i^* of different parameters.

The sensitivity coefficients for Manning's coefficient of pervious surface area (N-Perv) also showed a significant negative correlation, with the largest effect of -0.511 under the szzbl50a rain pattern, and -0.141 and -0.194 under the cxzd50a and Chicago50a rain patterns, respectively. Surface roughness (friction) also had a large effect on water volume, with sensitivity coefficients for all three rain patterns, especially the highest sensitivity coefficient of 0.316 under the Chicago50a rain pattern, indicating that an increase in surface roughness leads to more water accumulation. The velocity diffusion coefficient (Vd) showed a positive correlation in all three rainfall patterns, with sensitivity coefficients ranging from 0.155 to 0.185 , indicating that this parameter also has a significant effect on water volume.

Other parameters, such as decay coefficient (Dc), maximum infiltration rate (Maxrate), and ponding storage in the permeable zone (S-Perv), had relatively little effect on water volume. The sensitivity coefficient of decay time was 0.174 for cxzd50a and 0.048 and 0.026 for szsbl50a and Chicago50A, respectively. The sensitivity coefficient of maximum infiltration rate was negative for all three rainfall patterns, but the values were small, indicating that it had some effect on water volume reduction. The sensitivity coefficient of permeable area ponded storage is larger at -0.056 under the szzbl50a rainfall pattern, while the effect is smaller under the other two rainfall patterns.

Impervious area puddle storage (S-Imperv), stable infiltration rate (Fc), and Courant number (CFL) are less sensitive to water volume, and the sensitivity coefficients of these parameters are close to zero for all three rainfall patterns, indicating that they have a negligible effect on water volume. The order of sensitivity of different parameters is shown in Figure 4b.

4.2. Global Sensitivity Analysis

The global sensitivity analysis of the nonlinear relationship between the 12 parameters of the coupled model and the total amount of water accumulated in Honggu Tan District was conducted using the Sobol method, and Figure 5 shows the parameter sensitivities calculated by the Sobol method under different return periods. From Figure 5a, it can be seen that the S_1 values of Manning's coefficient of permeable water surface area (N-Perv) and minimum infiltration rate (Minrate) are significantly higher than the other parameters under different rainfall patterns, especially under the szzbl50a and Chicago50a rainfall patterns. Manning's coefficient of permeable surface area (N-Perv) had an S_1 value of 0.755 in szzbl50a, which was much higher than the other rainfall patterns, indicating that it was the most sensitive to the total amount of ponded water. Minimum infiltration rate (Minrate) reached an S_1 value of 0.684 for the cxzd50a rain pattern and 0.826 for the Chicago50a rain pattern, indicating that this parameter has a particularly significant effect on the total amount of standing water in these two rain patterns. In contrast, decay coefficient (Dc) is more moderately sensitive under the three rainfall patterns, with S_1 values ranging from 0.0278 to 0.057, showing some influence but much less than the Manning's coefficient for the permeable surface area (N-Perv) and the minimum infiltration rate (Minrate). In addition, pipe roughness (Roughness) showed a strong sensitivity under the Chicago50a rain type, with an S_1 value of 0.485, while it was significantly lower under the other rain types, indicating that the sensitivity of this parameter varies significantly between rain types. Comparing the values under the three rain patterns, it can be found that the sensitivity of Manning's coefficient of permeable surface area (N-Perv) and minimum infiltration rate (Minrate) is more prominent under cxzd50a and szzbl50a, indicating that they have a greater influence on the total amount of water accumulation. The sensitivity of minimum infiltration rate (Minrate) reached the highest value under the Chicago50a rainfall pattern, indicating that the contribution of Minrate to the total amount of standing water is significant under this rainfall pattern, followed by the Manning's coefficient of the permeable water surface area (N-Perv) and the stable infiltration rate (Fc), while the sensitivities of the other parameters, such as the Courant number (CFL) and the velocity diffusion coefficient (Vd), were less.

From the results in Figure 5b, it can be seen that the Manning coefficient (N-Perv) and the minimum infiltration rate (Minrate) of the permeable water surface area are the most significant parameters contributing to the total amount of standing water. The S_T values of N-Perv are 0.644 and 0.065 for the szzbl50a and Chicago50a rain patterns, respectively, indicating its dominance in the global influence of the system. And the S_T values of Minrate exceed 0.5 in both cxzd50a and Chicago50a, indicating that its sensitivity is also not negligible. In contrast, the very low values of Courant number (CFL) and impervious area puddle storage (S-Imperv) indicate that their global influence on water volume is negligible. These types of parameters have low sensitivity and relatively insignificant contributions to the system. The remaining parameters, such as surface roughness (Friction) and pipe roughness (Roughness), exhibit moderate global sensitivity under some rainfall patterns but have a relatively small effect on the total amount of standing water.

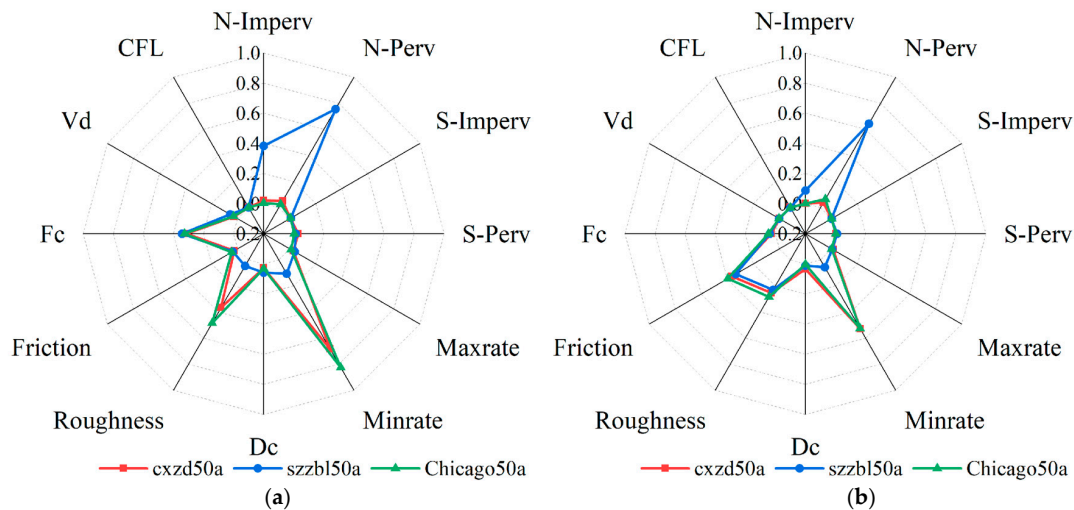


Figure 5. Sobol sensitivity analysis: (a) S_1 sensitivity; (b) S_T sensitivity.

In the sensitivity analysis, S_1 represents the first-order sensitivity of each parameter to the target output, i.e., the direct effect on the results when the parameter is varied individually. In contrast, S_T represents the global sensitivity, which includes the direct effect of the parameter and the interaction with other parameters. If $S_1 \approx S_T$, it indicates that the influence of the parameter comes mainly from the direct effect, and the interaction sensitivity is small, while if $S_1 < S_T$, it indicates that the parameter has significant interactions with other parameters, and the overall response of the system relies more on the synergistic influence among the parameters. Figure 6 illustrates the comparison between S_1 and S_T .

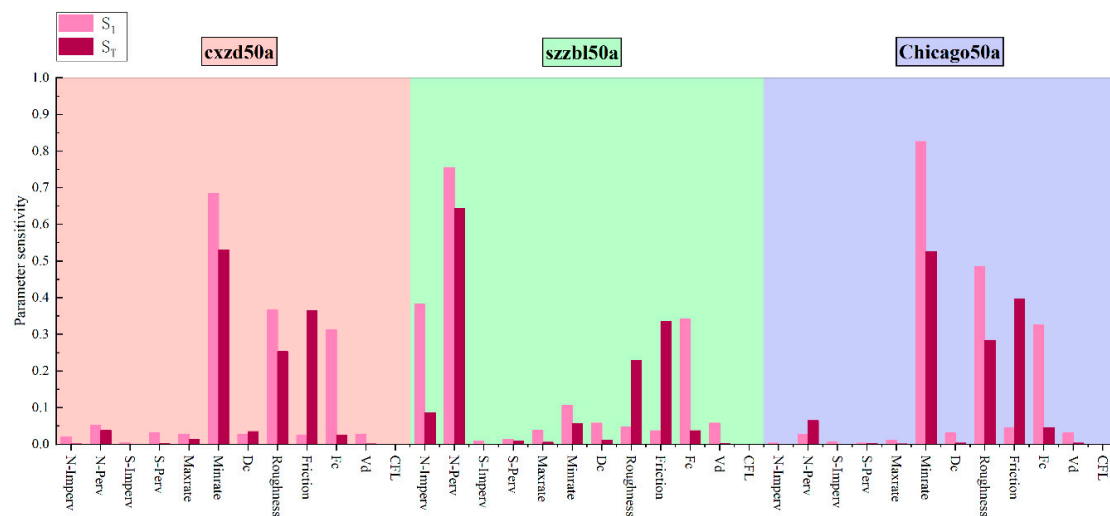


Figure 6. Comparison of Sobol's S_1 and S_T result.

A comparison of the results for S_1 and S_T reveals significant differences in the direct sensitivity and interaction of the key parameters under different rainfall patterns. To illustrate, the minimum infiltration rate (Minrate) is observed to have $S_1 > S_T$ in cxzd50a and Chicago50a, indicating that the effect of this parameter is predominantly manifested through the direct effect, with the interaction effect being relatively weak. In contrast, the interaction sensitivity of Minrate was more pronounced in szzbl50a. In the case of szzbl50a, the sum of Manning's coefficient of permeable surface area (N-Perv) exhibits a notable discrepancy of 0.76, which is considerably higher than the value of 0.64. This suggests that its direct impact is considerably more pronounced than its interaction with other variables. This may be attributed to the distribution of rainfall peaks, and further

investigation is necessary to gain a more comprehensive understanding. Similarly, pipe roughness (Roughness) exhibited analogous interaction characteristics. In Chicago50a and cxzd50a, the direct effect of Roughness is significant, as indicated by the greater value of S_1 compared to S_T . However, in szzbl50a, the effect is not as pronounced. However, the magnitude relationship between S_1 and S_T for surface roughness (friction) was consistently independent of the specific rain type, indicating that this parameter played a pivotal role in the interaction effect. In contrast, the extremely low and similar values of the sums of Courant number (CFL) and impervious area puddle storage (S-Imperv) indicate that these parameters have both low global and direct sensitivities to the system, little interaction sensitivity, and very limited effects. In general, the parameters that contribute most significantly to the total ponding volume are the minimum infiltration rate (Minrate) and surface roughness (Friction). However, other parameters, such as Manning's coefficient of permeable surface area (N-Perv) and pipeline roughness (Roughness), also have an effect under specific rainfall patterns.

5. Conclusions

This study employs a 1D–2D coupled waterlogging model for the Honggutun District of Nanchang City to conduct a sensitivity analysis of 12 parameters in the Storm Water Management Model and Telemac, using water volume as the objective function. The results indicate that minimum infiltration rate, Manning's coefficient of permeable surfaces, pipe roughness, and surface roughness have the most significant impact on total surface waterlogging volume, while other parameters exhibit varying degrees of sensitivity depending on rainfall patterns.

5.1. Identification of Key Sensitive Parameters

The analysis confirms that minimum infiltration rate and Manning's coefficient of permeable surfaces exert a dominant influence on surface water accumulation under all rainfall conditions. In contrast, pipe roughness and surface roughness show conditional sensitivity, depending on specific rainfall patterns. These findings emphasize the importance of carefully calibrating highly sensitive parameters to enhance model accuracy and predictive reliability.

5.2. Comparison of Morris and Sobol Methods for Sensitivity Analysis

The Morris method provides a computationally efficient screening tool to identify sensitive parameters and determine their influence on increasing or decreasing water volume. However, it does not quantify higher-order parameter interactions. The Sobol method, while computationally more demanding, offers a detailed variance-based assessment, allowing for a comprehensive evaluation of combined parameter effects. These findings highlight the complementary nature of both methods, supporting a hybrid approach to urban flood modeling that balances efficiency and analytical depth.

5.3. Impact of Rainfall Patterns on Parameter Sensitivity

Model parameter sensitivity varies significantly with different rainfall patterns. In cxzd50a and Chicago50a rainfall events, where precipitation intensifies early before tapering off, minimum infiltration rate and pipe roughness exhibit the highest sensitivity. In contrast, the szzbl50a rainfall pattern, characterized by two distinct peaks, results in greater sensitivity of the Manning's coefficient of impervious surfaces while reducing the influence of minimum infiltration rate and pipe roughness. This suggests that the temporal distribution of rainfall peaks plays a crucial role in parameter sensitivity, reinforcing the need for scenario-based calibration strategies in flood modeling.

5.4. Contextual Adaptation of Model Parameters

The model parameters in this study were empirically determined, ensuring objectivity and reliability. While certain geographic parameters (e.g., slope, catchment area, and land use) are region-specific and may limit direct applicability to other areas, the dataset used remains highly relevant for Nanchang, serving as a useful reference for urban flood modeling in this region. If applying this methodology elsewhere, sensitivity analyses should be adapted to local conditions to ensure model accuracy and reliability. This highlights the need for context-specific parameter calibration in hydrological modeling to improve adaptability across different urban settings.

5.5. Practical Implications for Urban Flood Management

These findings have significant implications for urban flood management and model optimization. The identification of key sensitive parameters aids in efficient model calibration, leading to more accurate flood predictions while reducing computational costs. Additionally, the variation in sensitivity across rainfall patterns underscores the necessity of scenario-based calibration strategies to improve model adaptability under different storm conditions.

The comparative assessment of the Morris and Sobol methods reinforces the advantages of hybrid sensitivity analysis, striking a balance between computational efficiency and analytical comprehensiveness, making it a practical choice for large-scale urban flood assessments. Integrating these insights into flood risk assessments and drainage system planning allows for urban water management agencies to prioritize critical parameters, improve early warning systems, and enhance resilience against extreme rainfall events.

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